Suspicious Sensor Analysis

Probability, Statistics, and Monte Carlo Infrastructure Study

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Notebook

`01_descriptive_statistics.ipynb`

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Environment

- Python 3.11+
- UV Environment

Purpose

This notebook explores pressure sensor data from a synthetic infrastructure monitoring system.

The analysis focuses on:

- descriptive statistics,
- probability distributions,
- z-score analysis,
- anomaly detection,
- Monte Carlo simulation,
- and engineering interpretation of operational uncertainty.

The objective is to develop practical understanding of probabilistic analysis techniques within an infrastructure engineering context.

Dataset

```
data/raw/pressure_sensor_readings.csv
```

Related Documentation

```
docs/00_overview/01_problem_statement.md
```

Topics Covered

- Mean
- Variance
- Standard deviation
- Probability distributions
- Z-scores
- Monte Carlo simulation
- Bayesian reasoning
- Infrastructure risk interpretation

Primary Libraries

```
 import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
```

Notes

This notebook is educational and analysis-focused.

Some methods prioritize readability and conceptual clarity over computational optimization.

Because understanding why a model behaves strangely is often more valuable than generating incorrect results with extreme confidence. Modern analytics occasionally forgets this.

```python
import numpy as np
import pandas as pd
from src.utils.descriptive_stats_functions import load_data
```

```python
df = load_data("pressure_sensor_readings.csv")
print(type(df))
```

```
<class 'pandas.DataFrame'>
```

```python
df.head(5)
```

	timestamp	pressure_psi	flow_gpm	sensor_status	pump_status	anomaly_flag
0	2026-01-01 00:00:00	73.74	1761.8	NORMAL	ON	0
1	2026-01-01 01:00:00	71.52	1576.5	NORMAL	ON	0
2	2026-01-01 02:00:00	74.28	1636.4	NORMAL	ON	0
3	2026-01-01 03:00:00	77.34	1597.6	NORMAL	ON	0
4	2026-01-01 04:00:00	71.20	1450.0	NORMAL	OFF	0

```python
df.shape
```

```
(1203, 6)
```

```python
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1203 entries, 0 to 1202
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   timestamp      1203 non-null   str    
 1   pressure_psi   1178 non-null   float64
 2   flow_gpm       1203 non-null   float64
 3   sensor_status  1203 non-null   str    
 4   pump_status    1203 non-null   str    
 5   anomaly_flag   1203 non-null   int64  
dtypes: float64(2), int64(1), str(3)
memory usage: 56.5 KB
```